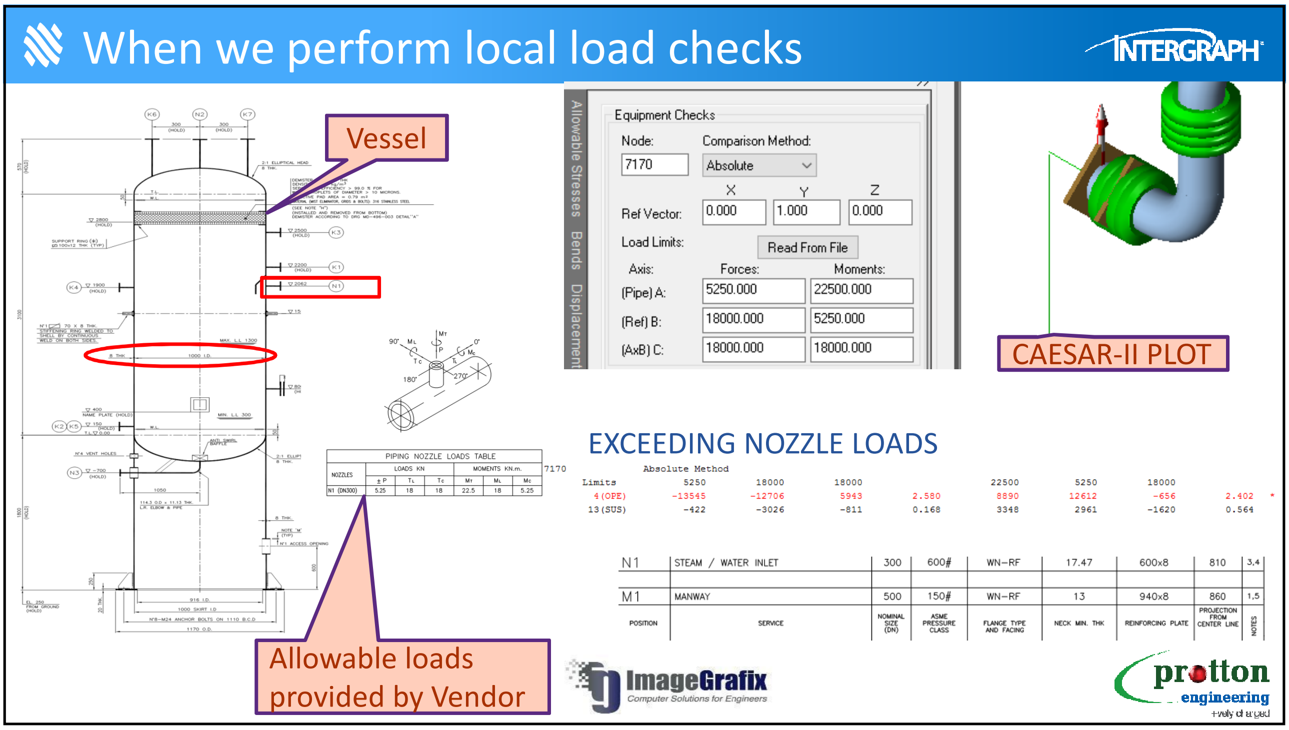

Presented by GAURAV BHENDE in CAU EXPRESS 2017

Methods to evaluate shell nozzle junction

INTERGRAPH

- PD 5500
- WRC 107(537)
- WRC 297
- FEA

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PD 5500

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PD5500(Annex G)

INTERGRAPH

- PD stands for Published Document no. 5500
- Formerly known as BS 5500:Specification for unfired, fusion welded pressure vessels
- Similar to WRC 107
- Withdrawn from the list of British Standards because it was not harmonized with the European Pressure Equipment Directive (97/23/EC)
- includes evaluation of nozzles on Head, whereas WRC addresses only cylinder-cylinder junction.

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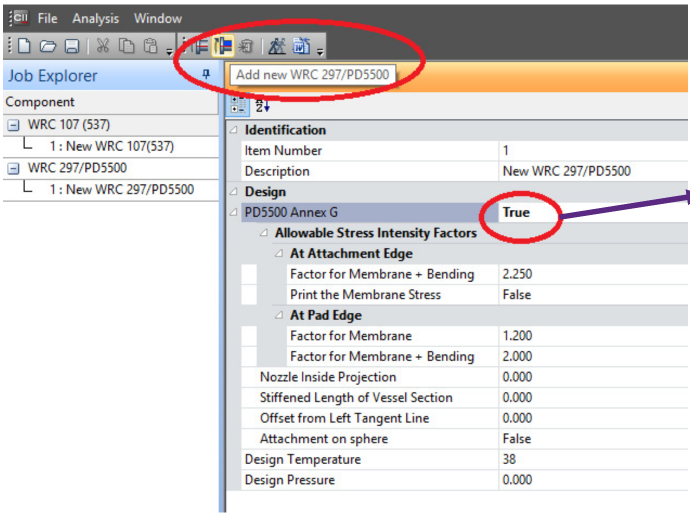
PD5500(Annex G)

INTERGRAPH



If this field is set to True, the program will compute local stresses in accordance with British Standard 5500 Annex G instead of WRC 297.

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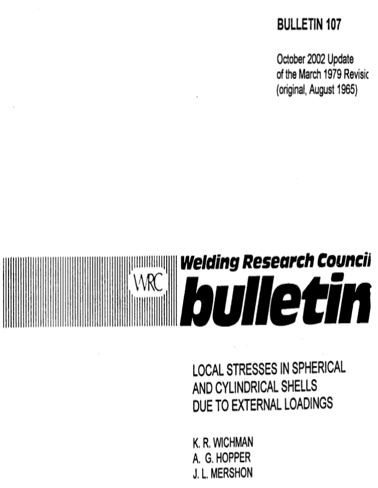
WRC 107

WRC 107(537)


BULLETIN 107

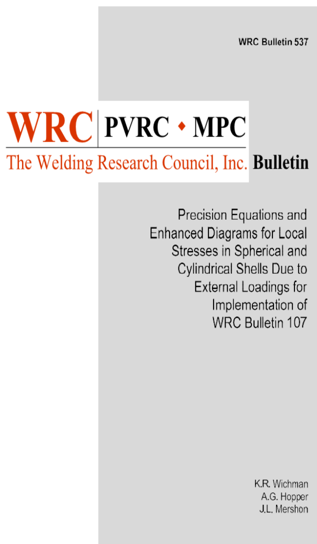
October 2002 Update
of the March 1979 Revisic
(original, August 1965)

Welding Research Council
bulletin

LOCAL STRESSES IN SPHERICAL
AND CYLINDRICAL SHELLS
DUE TO EXTERNAL LOADINGS

K. R. WICHMAN
A. G. HOPPER
J. L. MERSHON

- Welding Research Council Bulletin no. 107
- Based on research by Prof. P. P. Bijlaard in 1965 through research program by PVRC
- Now replaced by WRC 537
- WRC 537 Provides Precision Equations and Enhanced Diagrams
- useful and clear format


WRC Bulletin 537

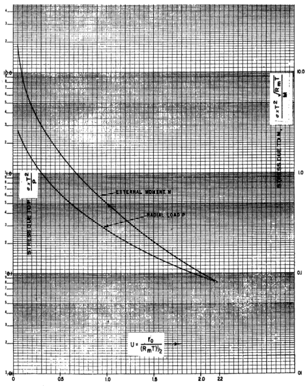
WRC | PVRC • MPC
The Welding Research Council, Inc. Bulletin

Precision Equations and
Enhanced Diagrams for Local
Stresses in Spherical and
Cylindrical Shells Due to
External Loadings for
Implementation of
WRC Bulletin 107

K.R. Wichman
A.G. Hopper
J.L. Mershon

WRC 107(537)

INTERGRAPH



- It is the 2010 printing of WRC 107.
- No longer delivers WRC 107 when requested for purchase
- Objective was to eliminate potential errors in implementation.

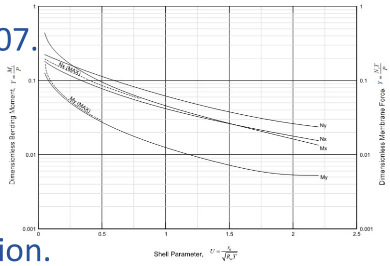


Figure SP-1 – Stresses in Spherical Shell Due to a Radial Load P on Nozzle Connection

- Dimensionless curves are represented by equations.

Curve Fit Coefficients for Figure SP-1

$$Y = \left(a + cU + eU^2 + gU^3 + iU^4\right) / \left(1 + bU + dU^2 + fU^3 + hU^4 + jU^5\right)$$

Coefficients	M_i			N_i		
	M_x	M_y	$M_{y(\max)}$	N_x	$N_{x(\max)}$	N_y
a	3.7231311E+00	1.9182958E+00	-5.3669486E+01	2.1322836E-01	2.1858968E-01	2.4586502E-01
b	1.8671976E+02	7.1931150E+02	-1.2516166E+04	4.2050448E+00	-2.5343227E+00	2.0008949E+02
c	2.6436830E+01	7.4335393E+01	2.2431247E+02	1.0916995E-01	-1.0771993E+00	4.8408477E+01
d	1.4194954E+03	2.6452748E+03	2.2926371E+05	5.4913965E+00	-3.6522378E+00	2.5345658E+02
e	6.4800756E+02	-4.5736180E+01	1.5781341E+04	1.4080117E+00	1.5900589E+00	-2.5034168E+01
f	9.7346418E+03	9.1388566E+02	1.0511173E+05	2.4926977E+01	1.1261041E+01	1.5866437E+01
g	-1.6034796E+02	-5.3218355E+00	4.7741439E+03	-2.4401880E-01	-4.6485921E-01	5.4387020E+00
h	0	-2.5983370E+03	2.2746897E+06	0	0	0
i	0	5.7654597E+00	2.2544280E+04	0	0	0
j	0	7.9658446E+02	8.0828856E+04	0	0	0

- Facilitates proper interpolation extrapolation.
- Permit efficient computation with modern computers

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SCOPE OF WRC107 (537)

INTERGRAPH

- Applicable for local stresses in Spherical and Cylindrical Shells.
- Not applicable for nozzle on conical end or elliptical dish end.
- Rectangular, Solid, Round, Hollow attachments can be analyzed.
- Based on un-penetrated shell.
- Considers Nozzle as rigid.

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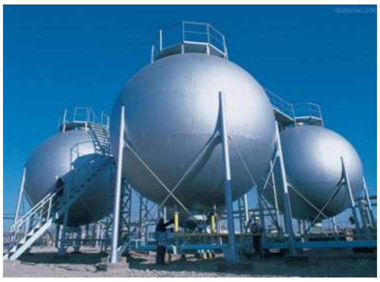
LIMITATIONS OF WRC107 (537)

INTERGRAPH

Geometrical limitations for Spherical shell:-

- $d_i/D_i \leq 0.33$ but less is D_m/T between 20-55.
- $U \leq 2.2$ (U is attachment parameter)
- $0.25 \leq t/T \leq 10$
- $5 \leq r_m/t \leq 50$



FOREWORD

ing surfaces for cylindrical vessels. These circumstances limited the potential usefulness of the results to d_i/D_i ratios of perhaps 0.33 in the case of spherical shells and 0.25 in the case of cylindrical shells. Since no data were available for the larger diameter ratios, Prof. Bijlaard later supplied data, at the urging of the design engineers, for the values of β = 0.375 and 0.50 (d_i/D_i ratios approaching 0.60) for cylindrical shells, as listed on page 12 of Reference 10. In so doing, Prof. Bijlaard in-

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LIMITATIONS OF WRC107 (537)

INTERGRAPH

Geometrical limitations for Cylindrical shell:-

- $d_i/D_i \leq 0.25$ or < 0.6 (with significant warning)
- $D/T \leq 600$
- $L/D \geq 1.5$
- $\beta \leq 0.5$ (β is shell parameter)
- $0.25 \leq C_1/C_2 \leq 4$ (C_1, C_2 are Half-length of rectangular loading in Circ./Long. direction, in)

FOREWORD

ing surfaces for cylindrical vessels. These circumstances limited the potential usefulness of the results to d_i/D_i ratios of perhaps 0.33 in the case of spherical shells and 0.25 in the case of cylindrical shells. Since no data were available for the larger diameter ratios, Prof. Bijlaard later supplied data, at the urging of the design engineers, for the values of β = 0.375 and 0.50 (d_i/D_i ratios approaching 0.60) for cylindrical shells, as listed on page 12 of Reference 10. In so doing, Prof. Bijlaard in-

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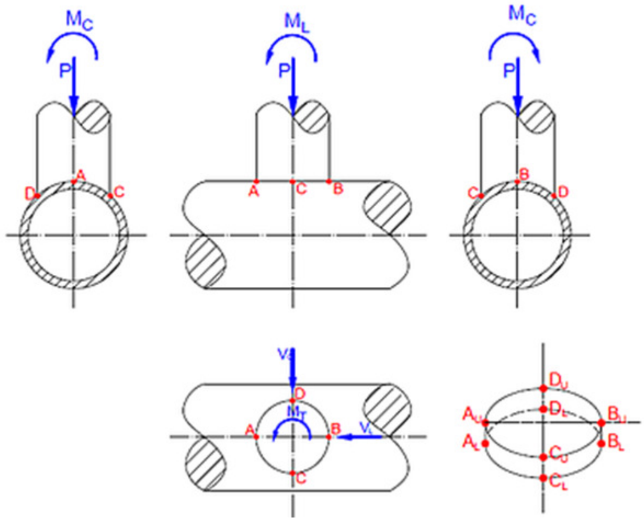
Nomenclature

INTERGRAPH



P:- Radial Load
V_L:- Longitudinal Shear
V_C:- Circumferential Shear
M_T:- Torsional Moment
M_C:- Circumferential Moment
M_L:- Longitudinal Moment

```
WRC 107 Stress Summations:

Vessel Stress Summation at Attachment Junction
------------------------------------------------------------------------------
  Type of      |                  Stress Values at
 Stress Int.   |                     (KPa          )
---------------|--------------------------------------------------------------
  Location     |   Au     Al     Bu     Bl     Cu     Cl     Du     Dl
---------------|--------------------------------------------------------------
```

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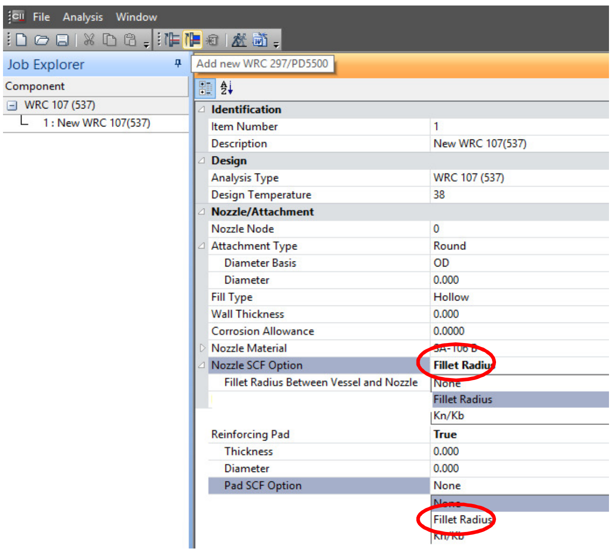
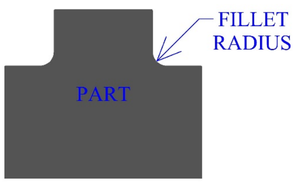
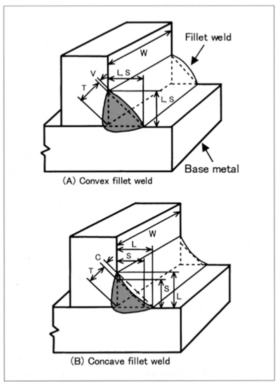
Input in CAESAR II:-Fillet Radius

INTERGRAPH



Fillet welding refers to the process of joining two pieces of metal together whether they be perpendicular or at an angle.





The default radius is equal to Radius = Throat*cos(45°)/(1-cos(45°)

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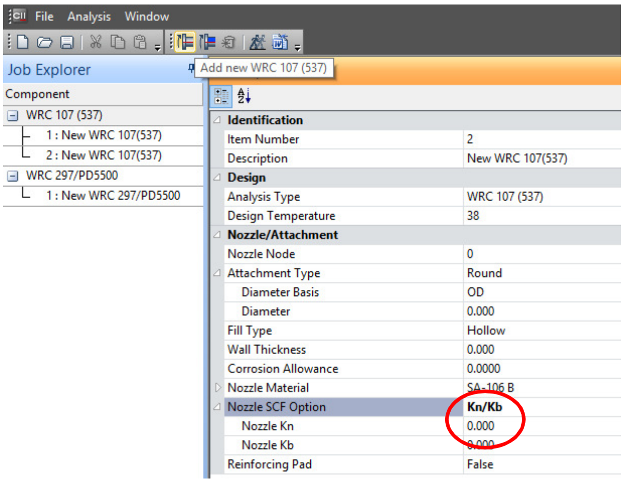
Input in CAESAR II:- Kn/Kb

INTERGRAPH



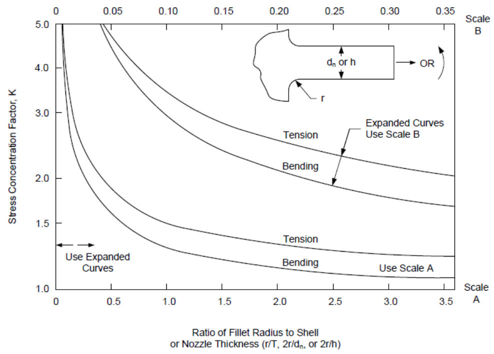


Figure B-2 – Stress Concentration Factors for $D >> d$

K_n – Membrane stress concentration factor
K_b – Bending stress concentration factor

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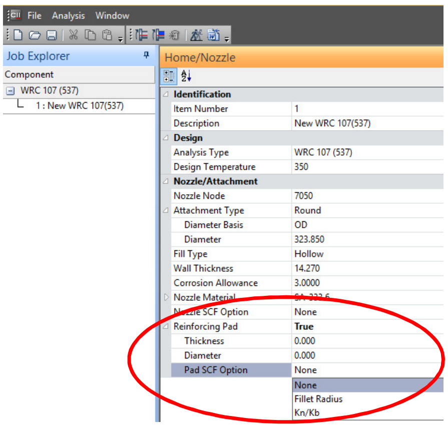
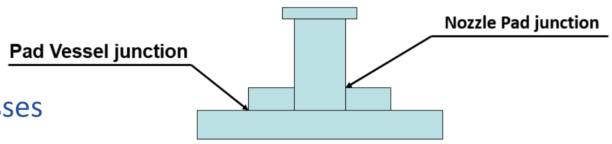
Input in CAESAR II:-Reinforcement Pad

INTERGRAPH



- WRC 107, 297 and 368: No separate calculation for Reinforcement pad.
- WRC368 recommends a rule of thumb that has been used successfully and provides somewhat accurate and generally conservative results.
- If $\text{Pad width} > 1.65 * \sqrt{RT}$ and $> \frac{d}{2}$ then the shell thickness can be increased by the amount of pad thickness.



When reinforcement pad is considered stresses are calculated independently at,

a) Nozzle pad junction
b) Pad vessel junction

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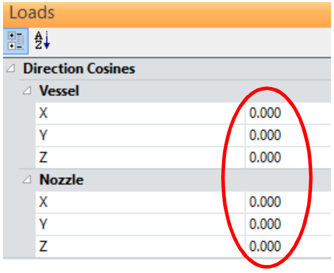
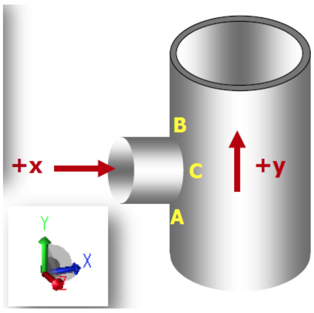
Input in CAESAR II:-Direction Cosine

INTERGRAPH

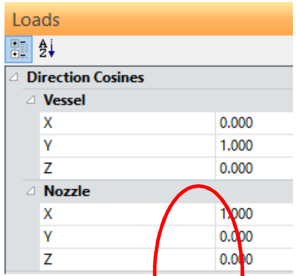


- The centerlines of the vessel and nozzle are required to be perpendicular to each other.
- The vessel direction vector is the vector pointing from the point B to A or A to B.
- The nozzle direction vector is vector pointing from the vessel nozzle connection to the centerline of vessel.



- The Vessel direction is +Y direction
- The Nozzle direction is +X direction

Direction cosines
Vessel are 0,1,0
Nozzle are 1,0,0



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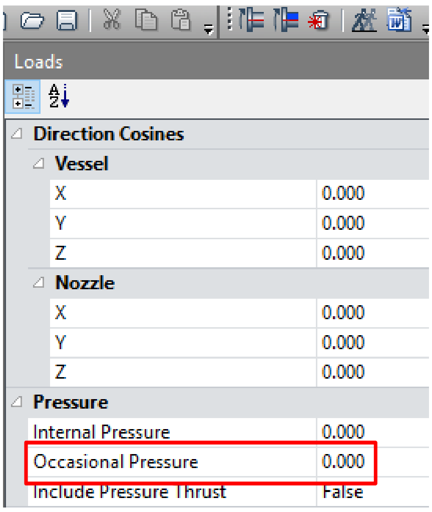
Input in CAESAR II:-OCC. Pressure

INTERGRAPH

Occasional Pressure (Pvar)



- P_{var} = DIFFERENCE between the peak pressure of the system and the system design pressure.
- Shall always be a positive (or 0) entry.
- Additional thrust load due to this pressure difference will also be accounted for in the nozzle radial loading in Pm(occ).
- This entry will be superimposed onto the system design pressure to evaluate the primary membrane stress due to occasional loads.
- Applicable only for WRC 107 analysis.

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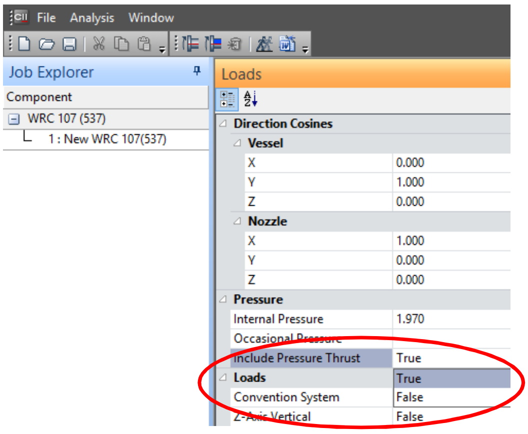
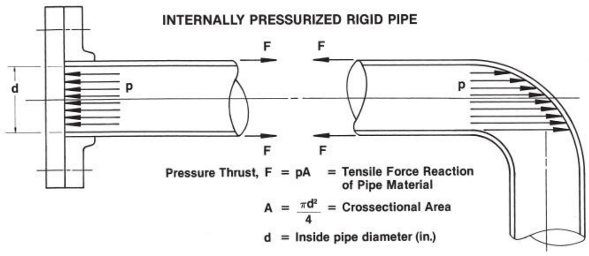
Input in CAESAR II:-Thrust Force

INTERGRAPH





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Input in CAESAR II:-Thrust Force

INTERGRAPH

- When 'include pressure thrust' option in WRC 107 module is 'checked', pressure thrust is added in radial force, only for sustained case

'include pressure thrust' : Unchecked

```
WRC 107 Stress Calculation for SUStained loads:
Radial Load                      P      -161.0  N.
Circumferential Shear            VC      -53.0  N.
Longitudinal Shear               VL    -2109.0  N.
Circumferential Moment           MC      121.0  N.m.
Longitudinal Moment              ML       33.0  N.m.
Torsional Moment                 MT     -775.0  N.m.
```

'include pressure thrust' : checked

```
WRC 107 Stress Calculation for SUStained loads:
Radial Load                      P   -135088.6  N.
Circumferential Shear            VC      -53.0  N.
Longitudinal Shear               VL    -2109.0  N.
Circumferential Moment           MC      121.0  N.m.
Longitudinal Moment              ML       33.0  N.m.
Torsional Moment                 MT     -775.0  N.m.
```

Calculation:-

THRUST FORCE(F)=PA= $\frac{P*\pi*d^2}{4} = \frac{1.97*\pi*(295.3)^2}{4}$ =134922 N.

TOTAL FORCE = Radial Load + Calculated Thrust Force=-161+ (- 134922)= -135083 N

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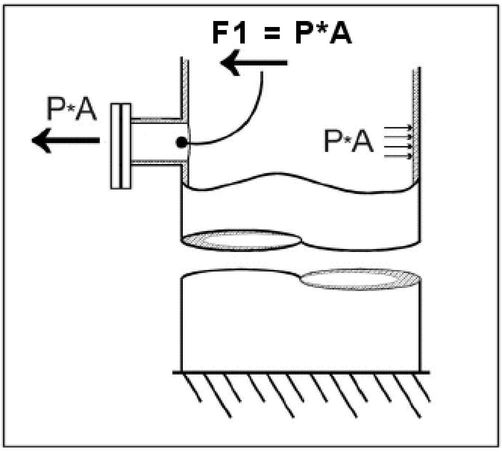
Presented by GAURAV BHENDE in CAU EXPRESS 2017

Input in CAESAR II:-Thrust Force

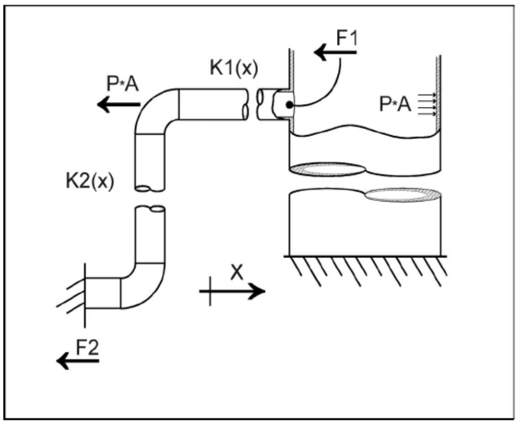
1. Vessel nozzle has a blind or blank flange.



- If the nozzle has a blind flange then it will experience the entire force due to the pressure thrust. We must include the whole pressure thrust load for this case.
- Hence, the amount of pressure thrust acting on a nozzle depends on the structural response of the system to a pressure load. If appropriate pressure thrust loads are applied to the piping and are analyzed, the structural load at the nozzle due to pressure can be calculated.

Input in CAESAR II:-Thrust Force

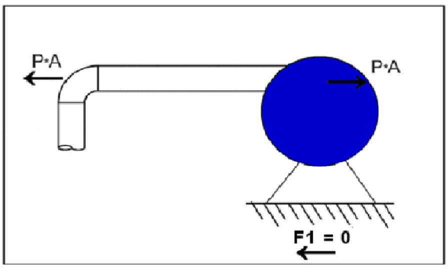
2. Straight run of pipe to an elbow, with no intermediate restraints.



- The pressure thrust load (PA) load acting on the elbow will be resolved at the nozzle and should be evaluated.

Input in CAESAR II:-Thrust Force

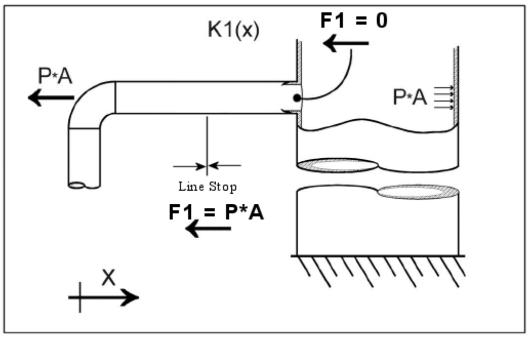
3. Straight run of 'rigid' pipe to a pump (or tied bellows unit in this line).



- The pressure thrust load (PA) load acting on the elbow will be balanced by PA load acting on the 'rigid' pump casing. Therefore F1= 0 resolved at the pump base an. Therefore, pressure thrust load should not be included in the nozzle evaluation.
- Note: If an un-tied axial bellows in the pump discharge line then pump base anchor will see the full PA. Both suction and discharge lines need to be examined.

Input in CAESAR II:-Thrust Force

4. Straight run of pipe to an elbow, with intermediate no-gap axial line stop restraint.

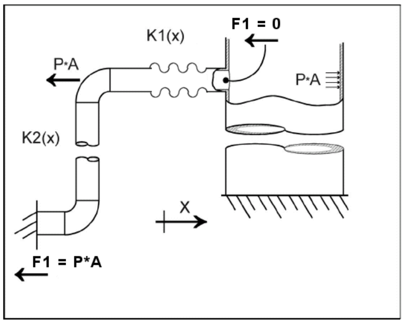


- The pressure thrust load (PA) load acting on the elbow will be resolved at the line stop. Pressure thrust load effect at the nozzle = 0.
- Therefore, pressure thrust load should not be included in the nozzle evaluation.

Input in CAESAR II:-Thrust Force

INTERGRAPH

5. Straight run of pipe to an elbow, with an intermediate untied axial expansion joint.



- The expansion joint is too flexible to transmit load, so the pressure thrust load (PA) will be resolved at restraints on the far side of elbow.
- Pressure thrust load effect at the nozzle = 0 or small value based on expansion stiffness and movement. Hence, pressure thrust load should not be included in the nozzle evaluation.
- If the expansion joint was tied or hinged, you would include the pressure thrust at the nozzle.

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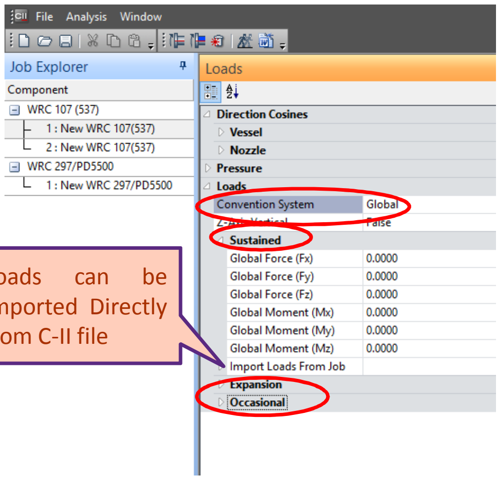
Input in CAESAR II:-Loads

INTERGRAPH



Loads can be Imported Directly from C-II file

Sustained load
- ❑ **Loads that are there for long periods**
- ❑ **Pressure**
- ❑ **Weight (on an attachments)**

Occasional
- ❑ **Loads that are momentary (lasting a short time)**
- ❑ **Wind loading**
- ❑ **Seismic loading**
- ❑ **Because they are short lived – add 20% to allowable stress**

Expansion
- ❑ **Strain controlled**
- ❑ **From thermal expansion of piping**
- ❑ **Always Secondary Stresses**

Operating loads
- ❑ **These are commonly Sustained + Thermal (eg: CAESAR II)**

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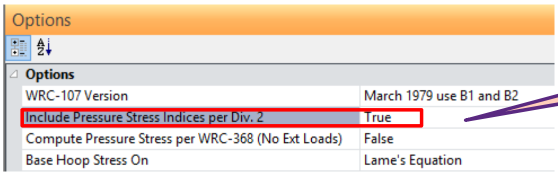
Input in CAESAR II:-Pressure Stress Indices

INTERGRAPH



Select True only for Fatigue Analysis

- Select 'True' to include the pressure stress indices described in ASME Sec. VIII Div. 2 Table AD-560.7.
- Estimates the peak stress intensity due to internal pressure.
- The peak stress intensity due to external loads is accounted by including the WRC 107 SIF (Kn and Kb factors).
- If we click stress indices, what we are computing is PL+Pb+Q+F.

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Input in CAESAR II:-Peak Stresses

INTERGRAPH

- Localized self-limiting stress.
- Exists at a discontinuity in the load path.
- Causes no objectionable distortion but it may be a possible source of fatigue failure.
- Computed by applying both the stress concentration factors and Pressure Stress Indices.
- Adds to primary and secondary stresses, to form the total stress.

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Input in CAESAR II:-WRC -368

- WRC-368, entitled "Stresses in Intersecting Cylinders Subjected to Pressure" was released in 1991.
- Calculates the stresses in cylinder to cylinder intersection (such as cylinder-nozzle), due to internal pressure and the pressure thrust loading.
- Calculates stresses in both the shell and the nozzle.
- Provides the maximum value of membrane stress intensity (general and local, Pm+PL) and the membrane + bending stress intensity (Pm+PL+Q).
- Provides a much more accurate modeling of the pressure thrust effect when the full thrust load acts on the vessel-nozzle junction such as in case of a nozzle with a blind.

Input in CAESAR II:-WRC -368

WRC 368 has geometric limitations:-

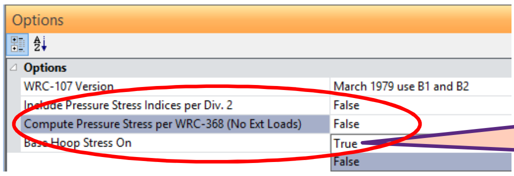
- 10 < D/T < 1000
- 4 < d/t < 1000
- 0.1 < t/T < 3
- 0.3 < Dt/dT < 6
- 0.3 < *d* / *Dt* < 6.5

```
WRC bulletin 368, Pressure Stress results:

Note: The following stresses are only due to internal pressure effects,
      including the pressure thrust loading. Other External loads are
      not considered.

Vessel Mean dia, D      :   1825.000 mm.
Vessel Corroded thk, T  :     19.000 mm.
Nozzle Mean dia, d      :    312.580 mm.
Nozzle Corroded thk, t  :     11.270 mm.
Design Pressure         :       1.97 N./sq.mm

Dimensionless parameters:
----------------------------------------------
Min Val    Param        Max Val       Value
----------------------------------------------
 10   <   D/T         < 1000   :     96.053
  4   <   d/t         < 1000   :     27.736
 .1   <   t/T         <    3   :      0.593
.03   <   d/D         <   .5   :      0.171
 .3   <   Dt/dT       <    6   :      3.463
 .3   <   d/(DT)^.5   <  6.5   :      1.679
```

CAESAR-II CHECKS WHETHER GEOMETRICAL LIMITITATION ARE SATISFIED OR NOT.

- Nozzle must be isolated (it may not be close to a discontinuity) – no two discontinuity shall be within 2.5 *RT* on vessel and not within 2.5 *rt* on nozzle.
- When these limits are exceeded then the results will not be as accurate.

Input in CAESAR II:-WRC -368



Select 'True' to compute pressure Stresses (in Shell and Nozzle) per WRC-368.

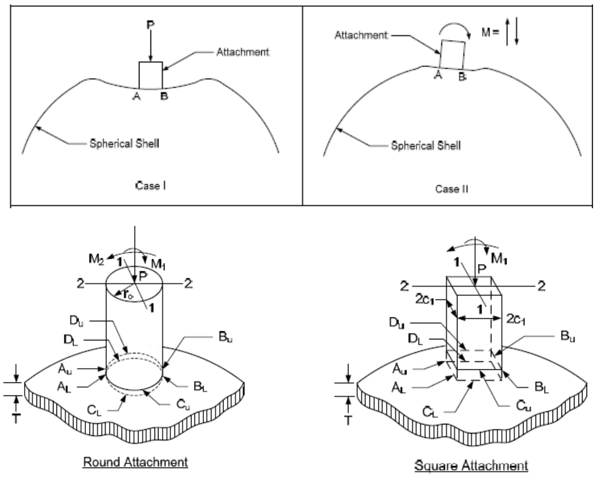
- WRC 368 conservative, in cases where only a portion of the thrust load acts on the nozzle as it considers thrust throughout the vessel
- There is no option to control the amount of thrust load.
- Does not provide information about the location and the orientation of the stresses.
- Utilizing WRC 368 along with WRC 107/297 is not very accurate for calculating the combined stress from pressure and external loads.

WRC 107 Analysis

Before starting the analysis

INTERGRAPH

Spherical Shell:-Sign Convention



P:-Radial Inward load
M:-Overturning Moment

Table 1 – Sign Convention for Stresses Resulting from Radial and Moment Load on a Spherical Shell

STRESS	LOCATION		LOADING P	LOADING M_1	LOADING M_2
Membrane $\frac{N_x}{T}$ & $\frac{N_y}{T}$	A_u	A_L	−		−
	B_u	B_L	−		+
	C_u	C_L	−	−	
	D_u	D_L	−	+	
Bending $\frac{6M_x}{T^2}$	A_u		−		−
	A_L		+		+
	B_u		−		+
	B_L		+		−
	C_u		−	−	
	C_L		+	+	
	D_u		−	+	
	D_L		+	−	
Bending $\frac{6M_y}{T^2}$	A_u		−		−
	A_L		+		+
	B_u		−		+
	B_L		+		−
	C_u		−	−	
	C_L		+	+	
	D_u		−	+	
	D_L		+	−	

Notes:
1. Sign convention for stresses: $(+)$ tension, $(-)$ compression.
2. If load P reverses, all signs in column P reverse
3. If overturning moment M_i reverses, all signs in column M_i reverse.
4. For round attachment, overturning moments M_1 and M_2 may be combined vectorially, thus: $M=\sqrt{M_1^2+M_2^2}$

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Before starting the analysis

INTERGRAPH

Cylindrical Shell:-Sign Convention

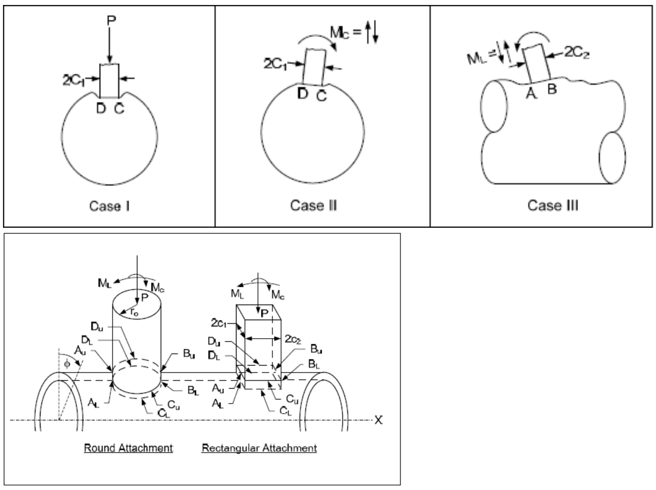

Table 4 Continued – Sign Convention and Location of Stresses

Table 4 – Sign Convention for Stresses Resulting from Radial and Moment Load on a Cylindrical Shell

STRESS	LOCATION		LOADING P	LOADING M_c	LOADING M_L
Membrane $\frac{N_\phi}{T}$ & $\frac{N_x}{T}$	A_u	A_L	−		−
	B_u	B_L	−		+
	C_u	C_L	−	−	
	D_u	D_L	−	+	
Bending $\frac{6M_x}{T^2}$	A_u		−		−
	A_L		+		+
	B_u		−		+
	B_L		+		−
	C_u		−	−	
	C_L		+	+	
	D_u		−	+	
	D_L		+	−	
Bending $\frac{6M_\phi}{T^2}$	A_u		−		−
	A_L		+		+
	B_u		−		+
	B_L		+		−
	C_u		−	−	
	C_L		+	+	
	D_u		−	+	
	D_L		+	−	

Notes:
1. Sign convention for stresses: $(+)$ tension, $(-)$ compression.
2. If load or moment directions reverse, all signs in applicable column reverse.

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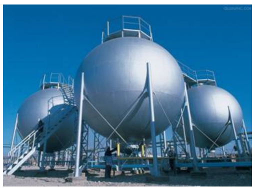
WRC 107(537) Theory:-Spherical Vs Cylindrical

INTERGRAPH



SPHERICAL SHELL

- Stresses will be covered in the vessel wall at the attachment –to-shell juncture.
- Shell Parameter (U) is the given by the ratio of the nozzle outside radius to the square root of the product of shell radius and thickness.

$$U = \frac{r_0}{\sqrt{R_m T}}$$

- Attachment Parameter For Hollow Cylindrical attachment of Spherical Shell

$$\rho = \frac{T}{t}$$

- For Hollow Square attachment of Spherical Shell

$$\tau = \frac{r_m}{0.875t} \qquad \rho = \frac{T}{t}$$

CYLINDRICAL SHELL

- Stresses will be considered in the shell at the attachment-to-shell juncture in both the circumferential and longitudinal directions.
- Shell Parameter (γ) is given by the ration of shell mid –radius to shell thickness.

$$\gamma = R_m / T$$

- Attachment Parameter For Hollow Round attachment of Cylindrical Shell

$$\beta = \frac{0.875\, r_0}{R_m}$$

- For Hollow Square attachment of Cylindrical Shell

$$\beta = \beta_1 = \beta_2 = \frac{c_1}{R_m} = \frac{c_2}{R_m}$$

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WRC 107(537) Theory:-Nomenclature

INTERGRAPH

Symbol	Represents
U	Shell parameter for spherical shell
γ	Shell parameter for cylindrical shell
R_m	Mean radius of cylindrical shell, in.
T	Wall thickness of cylindrical shell, in.
t	Thickness of hollow cylindrical attachment, in.
β	Attachment parameter
r_o	Outside radius of cylindrical attachment, in.
R_m	Mean radius of Spherical/Cylindrical shell, in.
r_m	Mean radius of hollow cylindrical attachment, in.
K_n	Membrane stress concentration factor
K_b	Bending stress concentration factor
r	Fillet Radius between Nozzle and Vessel, in.
Q	Secondary stress intensity

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WRC 107(537) Theory:-Nomenclature

Symbol	Represents
N_Φ, N_x	Membrane forces in shell wall in circumferential and longitudinal directions respectively with respect to the shell
M_Φ, M_x	Bending moments in shell wall in circumferential and longitudinal directions respectively with respect to the shell
P	Concentrated radial load, lb.
C_1	Half-length of rectangular loading in circumferential direction, in.
C_2	Half-length of rectangular loading in longitudinal direction, in.
V_C	Concentrated shear load in the circumferential direction, lb.
V_L	Concentrated shear load in the longitudinal direction, lb.
M_C	External overturning moment in circumferential direction with respect to the shell, in. lb.
M_L	External overturning moment in longitudinal direction with respect to the shell, in. lb.
M_T	Concentrated external torsional moment, in. lb.
P_m	General primary membrane stress intensity
P_l	Local primary membrane stress intensity
S_{mc}/S_{mh}	Vessel Cold/Hot Allowable Stress, lb. / sq.in

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WRC 107(537) Theory:-

- Non Dimensional Curves

Figure	Description
1A	Moment $\left(\frac{M_\phi}{M_c / R_m \beta}\right)$ due to M_c
2A	Moment $\frac{M_x}{M_c / R_m \beta}$ due to M_c
3A	Membrane force $\left(\frac{N_\phi}{M_c / R_m^2 \beta}\right)$ due to M_c
4A	Membrane force $\left(\frac{N_x}{M_c / R_m^2 \beta}\right)$ due to M_c
1B or 1B-1	Moment $\left(\frac{M_\phi}{M_L / R_m \beta}\right)$ due to M_L
2B or 2B-1	Moment $\frac{M_x}{M_L / R_m \beta}$ due to M_L
3B	Membrane force $\left(\frac{N_\phi}{M_L / R_m^2 \beta}\right)$ due to M_L
4B	Membrane force $\left(\frac{N_x}{M_L / R_m^2 \beta}\right)$ due to M_L
1C or 1C-1	Moment $\frac{M_\phi}{P}$ due to P
2C or 2C-1	Moment $\frac{M_x}{P}$ due to P
3C	Membrane force $\left(\frac{N_\phi}{P / R_m}\right)$ due to P
4C	Membrane force $\left(\frac{N_x}{P / R_m}\right)$ due to P

For **Hollow attachment(Nozzle)** of Cylindrical shell following curves are refereed based on the value of γ & β.

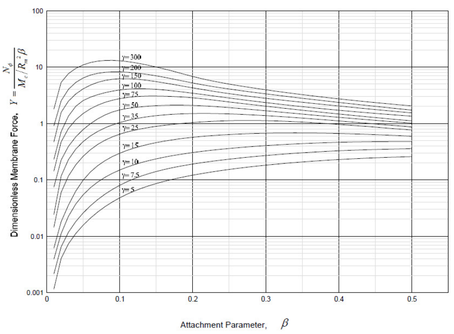

Figure 3A – Moment $\frac{N_\phi}{(M_c / R_m^2 \beta)}$ **Due to an External Circumferential Moment** M_c **on a Circular Cylinder –** ***Extrapolated***

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WRC 107(537) Theory:-

Computation sheet for local stresses in Cylindrical Shells(Hollow Attachment)

Table 5 – Computation Sheet for Local Stresses in Cylindrical Shells

1. Applied Loads	
Radial Load	$P =$
Circumferential Moment	$M_c =$
Longitudinal Moment	$M_L =$
Torsional Moment	$M_T =$
Shear Load	$V_c =$
Shear Load	$V_L =$
2. Geometry	
Vessel Thickness	$T =$
Attachment Radius	$r_o =$
Vessel Radius	$R_m =$
3. Geometric Parameters	
$\gamma = \frac{R_m}{T} =$	
$\beta = (0.875)\frac{r_o}{R_m} =$	
4. Stress Concentration Factors due to	
Membrane Load	$K_n =$
Bending Load	$K_b =$

Notes:
1. Enter all force values in accordance with sign convention.
2. Use consistent set of units in all calculations.

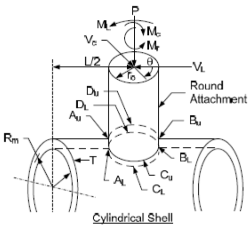

Table 5 Continued – Computation Sheet for Local Stresses in Cylindrical Shells

Reference Figure No.	Read Curves for	Calculate absolute values of stress and enter result	STRESSES - If load is opposite that shown, reverse signs shown							
			A_u	A_L	B_u	B_L	C_u	C_L	D_u	D_L
3C	$\frac{N_\phi}{P/R_m} =$	$K_n\left(\frac{N_\phi}{P/R_m}\right)\left(\frac{P}{R_m T}\right) =$	−	−	−	−	−	−	−	−
1C OR 2C-1	$\frac{M_\phi}{P} =$	$K_b\left(\frac{M_\phi}{P}\right)\left(\frac{6P}{T^2}\right) =$	−	+	−	+	−	+	−	+
3A	$\frac{N_\phi}{M_c/R_m^2\beta} =$	$K_n\left(\frac{N_\phi}{M_c/R_m^2\beta}\right)\left(\frac{M_c}{R_m^2\beta T}\right) =$					−	−	+	+
1A	$\frac{M_\phi}{M_c/R_m\beta} =$	$K_b\left(\frac{M_\phi}{M_c/R_m\beta}\right)\left(\frac{6M_c}{R_m\beta T^2}\right) =$					−	+	+	−
3B	$\frac{N_\phi}{M_L/R_m^2\beta} =$	$K_n\left(\frac{N_\phi}{M_L/R_m^2\beta}\right)\left(\frac{M_L}{R_m^2\beta T}\right) =$	−	−	+	+				
1B or 1B-1	$\frac{M_\phi}{M_L/R_m\beta} =$	$K_b\left(\frac{M_\phi}{M_L/R_m\beta}\right)\left(\frac{6M_L}{R_m\beta T^2}\right) =$	−	+	+	−				
Add algebraically for summation of ϕ stresses σ_ϕ		$\sigma_\phi =$								
4C	$\frac{N_x}{P/R_m} =$	$K_n\left(\frac{N_x}{P/R_m}\right)\left(\frac{P}{R_m T}\right) =$	−	−	−	−	−	−	−	−
1C-1 OR 2C	$\frac{M_x}{P} =$	$K_b\left(\frac{M_x}{P}\right)\left(\frac{6P}{T^2}\right) =$	−	+	−	+	−	+	−	+
4A	$\frac{N_x}{M_c/R_m^2\beta} =$	$K_n\left(\frac{N_x}{M_c/R_m^2\beta}\right)\left(\frac{M_c}{R_m^2\beta T}\right) =$					−	−	+	+
2A	$\frac{M_x}{M_c/R_m\beta} =$	$K_b\left(\frac{M_x}{M_c/R_m\beta}\right)\left(\frac{6M_c}{R_m\beta T^2}\right) =$					−	+	+	−
4B	$\frac{N_x}{M_L/R_m^2\beta} =$	$K_n\left(\frac{N_x}{M_L/R_m^2\beta}\right)\left(\frac{M_L}{R_m^2\beta T}\right) =$	−	−	+	+				
2B or 2B-1	$\frac{M_x}{M_L/R_m\beta} =$	$K_b\left(\frac{M_x}{M_L/R_m\beta}\right)\left(\frac{6M_L}{R_m\beta T^2}\right) =$	−	+	+	−				
Add algebraically for summation of x stresses σ_x		$\sigma_x =$								
Shear stress due to Torsion, M_T		$\tau_{\phi x} = \tau_{x\phi} = \frac{M_T}{2\pi r_o^2 T} =$	+	+	+	+	+	+	+	+
Shear stress due to load V_C		$\tau_{x\phi} = \frac{V_C}{\pi r_o T} =$	+	+	−	−				
Shear stress due to load V_L		$\tau_{x\phi} = \frac{V_L}{\pi r_o T} =$					−	−	+	+
Add algebraically for summation of shear stresses τ		$\tau =$								

COMBINED STRESS INTENSITY - S

1) When σ_ϕ and σ_x have like signs $S = \frac{1}{2}\left[\sigma_\phi + \sigma_x \pm \sqrt{(\sigma_\phi - \sigma_x)^2 + 4\tau^2}\right]$ or $\sqrt{(\sigma_\phi - \sigma_x)^2 + 4\tau^2}$
2) When $\tau = 0$, $S =$ largest absolute magnitude of either $S = \sigma_\phi, \sigma_x$, or $|\sigma_\phi - \sigma_x|$
3) When σ_ϕ and σ_x have unlike signs, $S = \sqrt{(\sigma_\phi - \sigma_x)^2 + 4\tau^2}$

Types of Analysis:-Elastic Analysis

If an elastic analysis is the only requirement, then the following points should be kept in mind when using the WRC 107 module:

- Set up different types of load cases (sustained, expansion and occasional).
- Include pressure thrust if needed.
- Do not include pressure stress indices if fatigue analysis is not an objective.
- Do not include stress concentration factors Kn and Kb (this is done by omitting the fillet radius between the vessel and nozzle).
- Use stress summation to compare the actual stress to the allowable stress.

Types of Analysis:- Fatigue Analysis

INTERGRAPH

If Fatigue analysis is the requirement, then the following points should be kept in mind when using the WRC 107 module:

- ASME VIII Div.2 paragraph AD-160 should be checked to see if the fatigue effect needs to be considered.
- When a fatigue evaluation is required, ASME Sec VIII, Div. 2, Appendix-4 and Appendix-5 should be used together.
- Elastic Analysis need to be performed before fatigue analysis.
- Include Peak stress intensities, Pressure Thrust.
- Include SCF Kn and Kb, by entering the fillet radius between the vessel and nozzle. Doing so will compute the peak stress due to applied loads.
- CAESAR-II currently DOES NOT perform the complete fatigue analysis per Section VIII Div.2 Appendix 4 & 5 rules.

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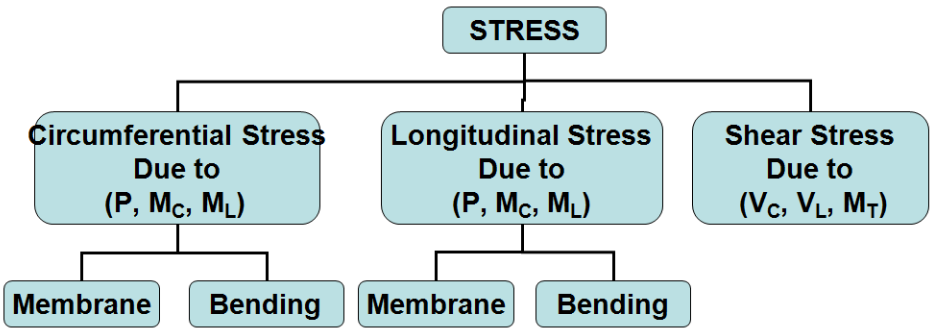
Types of Stresses:-

INTERGRAPH

Types of Stresses as per WRC107



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Stress Classification:-

INTERGRAPH

- The stresses calculated at the eight points, in the form of circumferential, longitudinal and shear stress are further combined as per the definitions given in ASME Section VIII Div. 2 in order to compare the summation stresses against suitable allowable.
- WRC 107 does not have any reference to allowable stresses and the same are taken from ASME Section VIII Div. 2

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Stress Classification:-

INTERGRAPH

- Stress
  - Primary stress
    - General primary Membrane stress intensity (P_m)
      - Due to
        - Internal pressure
    - Local primary Membrane stress intensity (P_l)
      - Due to
        - Membrane stress due to applied sustained forces and moment
  - Secondary stress
    - (Q)
      - Due to
        - Bending stress due to applied sustained forces and moments
        - Membrane and bending stress due to applied expansion forces and moments

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Stress Classification:-

Calculation of Vessel Stresses:-

- ASME Sec VIII, Div. 2, Appendix-4 states that the following limits must be satisfied.

$P_m < kS_{mh}$

$P_m + P_l + P_b < 1.5kS_{mh}$

$P_m + P_l + P_b + Q < 3S_{mavg}$

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Stress Classification:-

- As per the stress classification defined by Section VIII, Div. 2, the bending stress terms caused by any external load moments or internal pressure in the vessel wall near a nozzle, should be classified as Q, or the secondary stresses, regardless of whether they were caused by sustained or expansion loads. This causes P_b to disappear, and three criteria becomes more simple as:

$P_m < kSmh$

$P_m + P_1 < 1.5kSmh$

$P_m + P_1 + Q < 3Smavg$

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Stress Classification:-

INTERGRAPH

P_m → General primary membrane stress (primarily due to internal pressure)

P_l → Local primary membrane stress, which may include
--Membrane stress due to internal pressure
--Local membrane stress due to applied sustained forces and moments

Q → Secondary stresses, which may include
--Bending stress due to internal pressure
--Bending stress due to applied sustained forces & moments
--Membrane stress due to applied expansion forces
--Bending stress due to applied expansion forces & moments
--Membrane stress due to applied expansion moments

K → Occasional Stress factor

S_{mh} → Hot material allowable stress intensity

S_{mc} → Cold material allowable stress intensity

S_{mavg} → Average material stress intensity $(S_{mh} + S_{mc})/2$

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WRC-107 WARNING

INTERGRAPH

- While using WRC 107 there are no specific geometric restrictions available, hence it is the responsibility of the user to take care that the curve values are not exceeded

•The output report notifies the user if the curve values are exceeded with an exclamation (!) mark and warning printed in blue.

•If the curve value is exceeded, one should opt for Nozzlepro.

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979 B1/B2	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.155	4C	7.955	(A,B)
N(PHI) / (P/Rm)	0.155	3C	5.663	(C,D)
M(PHI) / (P)	0.155	2C1	0.042	(A,B)
M(PHI) / (P)	0.155	1C !	0.076	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.155	3A	2.167	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.155	1A	0.078	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.155	3B	5.742	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.155	1B1	0.029	(A,B,C,D)
N(x) / (P/Rm)	0.155	3C	5.663	(A,B)
N(x) / (P/Rm)	0.155	4C	7.955	(C,D)
M(x) / (P)	0.155	1C1	0.078	(A,B)
M(x) / (P)	0.155	2C !	0.043	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.155	4A	3.831	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.155	2A	0.039	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.155	4B	2.105	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.155	2B1	0.041	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

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WRC ANALYSIS:-Output Report Study

Input Echo, 107/537 Item 1, Description: New WRC 107(537

Diameter Basis for Vessel	Vbasis	OD	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	1844.000	mm.
Vessel Thickness	Tv	22.000	mm.
Vessel Node Number		7051	
Design Temperature		350.00	C
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	137892.00	KPa
Vessel Hot S.I. Allowable	Smh	128465.29	KPa
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	323.850	mm.
Nozzle Thickness	Tn	14.270	mm.
Nozzle Material		SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	117897.66	KPa
Nozzle Hot S.I. Allowable	SNmh	115426.55	KPa
Nozzle Node Number		7050	
Design Internal Pressure	Dp	1.970	N./sq.mm.
Include Pressure Thrust		No	

Vessel Input Data

Nozzle Input Data

C

Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	
Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	
Global Force (SUS)	Fx	-161.0	N.
Global Force (SUS)	Fy	-2109.0	N.
Global Force (SUS)	Fz	53.0	N.
Global Moment (SUS)	Mx	775.0	N.m.
Global Moment (SUS)	My	-121.0	N.m.
Global Moment (SUS)	Mz	-33.0	N.m.
Internal Pressure (SUS)	P	1.97	N./sq.mm.
Include Pressure Thrust		No	
Global Force (EXP)	Fx	1085.0	N.
Global Force (EXP)	Fy	-1636.0	N.
Global Force (EXP)	Fz	-4936.0	N.
Global Moment (EXP)	Mx	-8657.0	N.m.
Global Moment (EXP)	My	-3933.0	N.m.
Global Moment (EXP)	Mz	-3031.0	N.m.
Global Force (OCC)	Fx	1162.0	N.
Global Force (OCC)	Fy	-330.0	N.
Global Force (OCC)	Fz	-107.0	N.
Global Moment (OCC)	Mx	416.0	N.m.
Global Moment (OCC)	My	1175.0	N.m.
Global Moment (OCC)	Mz	-1486.0	N.m.
Occasional Internal Pressure (OCC)	Pvar	0.00	N./sq.mm.

Direction Cosine

Imported SUS Loads

Imported EXP Loads

Imported OCC Loads

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WRC ANALYSIS:-Output Report Study

INTERGRAPH

```
Occasional Internal Pressure (OCC)          Pvar        0.00  N./sq.mm.

Use Interactive Control                               No
WRC107 Version                           Version  March 1979 ( B1 & B2 )

Include Pressure Stress Indices per Div. 2               No
Compute Pressure Stress per WRC-368                      No
```

Note:
WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".

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WRC ANALYSIS:-Output Report Study

INTERGRAPH

```
WRC 107 Stress Calculation for SUStained loads:
Radial Load                                  P        -161.0  N.
Circumferential Shear                        VC        -53.0  N.
Longitudinal Shear                           VL      -2109.0  N.
Circumferential Moment                       MC        121.0  N.m.
Longitudinal Moment                          ML         33.0  N.m.
Torsional Moment                             MT       -775.0  N.m.

Dimensionless Parameters used :   Gamma =    48.03

Dimensionless Loads for Cylindrical Shells at Attachment Junction:
-------------------------------------------------------------------
Curves read for 1979 B1/B2          Beta   Figure     Value    Location
-------------------------------------------------------------------
N(PHI) / ( P/Rm )                   0.155   4C        7.273     (A,B)
N(PHI) / ( P/Rm )                   0.155   3C        5.343     (C,D)
M(PHI) / ( P )                      0.155   2C1       0.046     (A,B)
```

Program will Convert the Loads from Global Convention to WRC-107 convention

Gamma(γ)=$\frac{R_m}{T}$ = $\frac{922+900}{19}$ = 47.94

Beta(β)=$\frac{0.875*r_0}{R_m}$ = $\frac{0.875*162}{911}$ = 0.155

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WRC ANALYSIS:-Output Report Study

INTERGRAPH

```
Dimensionless Parameters used :   Gamma =   48.03

Dimensionless Loads for Cylindrical Shells at Attachment Junction:
-------------------------------------------------------------------
Curves read for 1979 B1/B2          Beta  Figure     Value   Location
-------------------------------------------------------------------
N(PHI) / ( P/Rm )                   0.155  4C        7.273   (A,B)
N(PHI) / ( P/Rm )                   0.155  3C        5.343   (C,D)
M(PHI) / ( P )                      0.155  2C1       0.046   (A,B)
M(PHI) / ( P )                      0.155  1C        0.079   (C,D)
N(PHI) / ( MC/(Rm**2 * Beta) )      0.155  3A        1.935   (A,B,C,D)
M(PHI) / ( MC/(Rm    * Beta) )      0.155  1A        0.080   (A,B,C,D)
N(PHI) / ( ML/(Rm**2 * Beta) )      0.155  3B        5.217   (A,B,C,D)
M(PHI) / ( ML/(Rm    * Beta) )      0.155  1B1       0.031   (A,B,C,D)

N(x)   / ( P/Rm )                   0.155  3C        5.343   (A,B)
N(x)   / ( P/Rm )                   0.155  4C        7.273   (C,D)
M(x)   / ( P )                      0.155  1C1       0.082   (A,B)
M(x)   / ( P )                      0.155  2C        0.045   (C,D)
N(x)   / ( MC/(Rm**2 * Beta) )      0.155  4A        3.318   (A,B,C,D)
M(x)   / ( MC/(Rm    * Beta) )      0.155  2A        0.040   (A,B,C,D)
N(x)   / ( ML/(Rm**2 * Beta) )      0.155  4B        1.861   (A,B,C,D)
M(x)   / ( ML/(Rm    * Beta) )      0.155  2B1       0.045   (A,B,C,D)
```

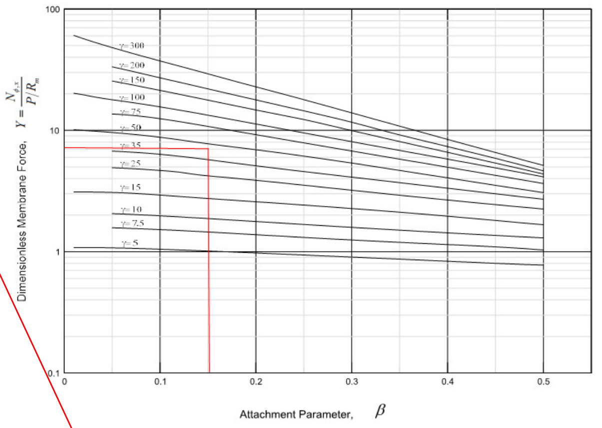

Figure 4C – Membrane Force $N_x/(P/R_m)$ Due to an External Radial Load P on a Circular Cylinder (Transverse Axis)
Membrane Force $N_\phi/(P/R_m)$ Due to an External Radial Load P on a Circular Cylinder (Longitudinal Axis) – *Original*

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INTERGRAPH

Stress Concentration Factors Kn = 1.00,   Kb = 1.00

Stresses in the Vessel at the Attachment Junction

Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
		Stress Values at (KPa)							
Circ. Memb.	P	67	67	67	67	49	49	49	49
Circ. Bend.	P	121	-121	121	-121	210	-210	210	-210
Circ. Memb.	MC	0	0	0	0	-95	-95	95	95
Circ. Bend.	MC	0	0	0	0	-1138	1138	1138	-1138
Circ. Memb.	ML	-70	-70	70	70	0	0	0	0
Circ. Bend.	ML	-118	118	118	-118	0	0	0	0
Tot. Circ. Str.		0	-5	378	-102	-973	882	1493	-1204

Circ. Memb. Stress (P) = $K_n \left(\frac{N\text{ø}}{P/Rm}\right) \left(\frac{P}{Rm * T}\right)$

$= (7.273)\left(\frac{161}{911 * 19}\right) * 1000 = 67.65$ kPa

Circ. Bend. Stress (P) = $K_b \left(\frac{M\text{ø}}{P}\right) \left(\frac{6P}{T^2}\right)$

$= (0.046)\left(\frac{6*161}{19^2}\right) * 1000 = 123.09$ kPa

Circ. Memb. Stress (MC) = $K_n \left(\frac{N\text{ø}}{M_c/R^2{}_m\beta}\right) \left(\frac{M_c}{R^2{}_m\beta * T}\right)$

$= (1.935)\left(\frac{121*1000}{911^2 * 0.155*19}\right) * 1000 = 95.79$ kPa

Circ. Memb. Stress (ML) = $K_n \left(\frac{N_\text{ø}}{M_c/R^2{}_m\beta}\right) \left(\frac{M_L}{R^2{}_m\beta * T}\right)$

$= (5.217)\left(\frac{33*1000}{911^2 * 0.155*19}\right) * 1000 = 70.43$ kPa

Circ. Bend. Stress (MC) = $K_b \left(\frac{M\text{ø}}{M_c/Rm\beta}\right) \left(\frac{6Mc}{R_m\beta * T^2}\right)$

$= (0.080)\left(\frac{6*121*1000}{911 * 0.155*19^2}\right) * 1000 = 1139$ kPa

Circ. Bend. Stress (ML) = $K_b \left(\frac{M_\text{ø}}{M_c/Rm\beta}\right) \left(\frac{6ML}{R_m\beta * T^2}\right)$

$= (0.031)\left(\frac{6*33*1000}{911 * 0.155*19^2}\right) * 1000 = 120.41$ kPa

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Stresses in the Vessel at the Attachment Junction

Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
		Stress Values at (KPa)							
Long. Memb.	P	49	49	49	49	67	67	67	67
Long. Bend.	P	219	-219	219	-219	120	-120	120	-120
Long. Memb.	MC	0	0	0	0	-163	-163	163	163
Long. Bend.	MC	0	0	0	0	-571	571	571	-571
Long. Memb.	ML	-24	-24	24	24	0	0	0	0
Long. Bend.	ML	-172	172	172	-172	0	0	0	0
Tot. Long. Str.		71	-22	466	-317	-545	354	923	-461

Circ. Memb. Stress (P) = $K_n(\frac{N_x}{P/Rm})\ (\frac{P}{Rm*T})$

$= (5.343)\ (\frac{161}{911*19})$ *1000= 49.70 kPa

Circ. Bend. Stress (P) = $K_b(\frac{M_x}{P})\ (\frac{6P}{T^2})$

$= (0.082)\ (\frac{6*161}{19^2})$ *1000= 219.42 kPa

Circ. Memb. Stress (MC) = $K_n(\frac{N_x}{M_c/R^2{}_m\beta})\ (\frac{M_c}{R^2{}_m\beta*T})$

$= (3.318)\ (\frac{121*1000}{911^2*0.155*19})$ *1000=164.26 kPa

Circ. Memb. Stress (ML) = $K_n(\frac{N_x}{M_c/R^2{}_m\beta})\ (\frac{M_L}{R^2{}_m\beta*T})$

$= (1.861)(\frac{33*1000}{911^2*0.155*19})$ *1000= 25.12 kPa

Circ. Bend. Stress (MC) = $K_b(\frac{M_x}{M_c/Rm\beta})\ (\frac{6Mc}{R_m\beta*T^2})$

$= (0.040)\ (\frac{6*121*1000}{911*0.155*19^2})$ *1000= 569.69 kPa

Circ. Bend. Stress (ML) = $K_b(\frac{M_x}{M_c/Rm\beta})\ (\frac{6ML}{R_m\beta*T^2})$

$= (0.045)\ (\frac{6*33*1000}{911*0.155*19^2})$ *1000= 174.79 kPa

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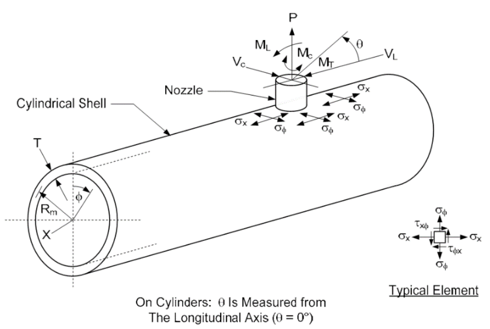
INTERGRAPH

Stresses in the Vessel at the Attachment Junction

Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
		Stress Values at (KPa)							
Shear	VC	-5	-5	5	5	0	0	0	0
Shear	VL	0	0	0	0	218	218	-218	-218
Shear	MT	-247	-247	-247	-247	-247	-247	-247	-247
Tot. Shear		-253	-253	-242	-242	-29	-29	-465	-465



$$\text{Shear } V_C = \frac{V_C}{\text{л} r_0 T}\cos(\Theta)$$

$$= \frac{53}{\text{л}*162*19} * \cos(0)*1000 \quad = 5.480 \text{ kPa}$$

$$\text{Shear } V_L = \frac{V_L}{\text{л} r_0 T}\sin(\Theta)$$

$$= \frac{2109}{\text{л}*162*19} * \sin(0)*1000 \quad = 218.10 \text{ kPa}$$

$$\text{Shear } M_T = \frac{M_T}{2\text{л} {r_0}^2 T}$$

$$= (\frac{775*1000}{2\text{л}*162^2*19}) * 1000 = 247.36 \text{ kPa}$$

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INTERGRAPH

Stresses in the Vessel at the Attachment Junction

Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
		Stress Values at (KPa)							
Circ. Memb.	P	67	67	67	67	49	49	49	49
Circ. Bend.	P	121	-121	121	-121	210	-210	210	-210
Circ. Memb.	MC	0	0	0	0	-95	-95	95	95
Circ. Bend.	MC	0	0	0	0	-1138	1138	1138	-1138
Circ. Memb.	ML	-70	-70	70	70	0	0	0	0
Circ. Bend.	ML	-118	118	118	-118	0	0	0	0
Tot. Circ. Str. σ_ϕ		0	-5	378	-102	-973	882	1493	-1204
Long. Memb.	P	49	49	49	49	67	67	67	67
Long. Bend.	P	219	-219	219	-219	120	-120	120	-120
Long. Memb.	MC	0	0	0	0	-163	-163	163	163
Long. Bend.	MC	0	0	0	0	-571	571	571	-571
Long. Memb.	ML	-24	-24	24	24	0	0	0	0
Long. Bend.	ML	-172	172	172	-172	0	0	0	0
Tot. Long. Str. σ_X		71	-22	466	-317	-545	354	923	-461
Shear	VC	-5	-5	5	5	0	0	0	0
Shear	VL	0	0	0	0	218	218	-218	-218
Shear	MT	-247	-247	-247	-247	-247	-247	-247	-247
Tot. Shear τ		-253	-253	-242	-242	-29	-29	-465	-465
Str. Int.		511	506	668	529	975	883	1754	1428

COMBINED STRESS INTENSITY - S

1) When σ_ϕ and σ_x have like signs $S = \frac{1}{2}\left[\sigma_\phi + \sigma_x \pm \sqrt{(\sigma_\phi - \sigma_x)^2 + 4\tau^2}\right]$ or $\sqrt{(\sigma_\phi - \sigma_x)^2 + 4\tau^2}$

2) When $\tau = 0$, $S =$ largest absolute magnitude of either $S = \sigma_\phi, \sigma_x$, or $|\sigma_\phi - \sigma_x|$

3) When σ_ϕ and σ_x have unlike signs, $S = \sqrt{(\sigma_\phi - \sigma_x)^2 + 4\tau^2}$

$$S = \sqrt{(\sigma_\phi - \sigma_X)^2 + 4\tau^2}$$
$$= \sqrt{(0 - 71)^2 + 4(253)^2}$$
$$= 510.95$$

- Similarly the stresses are calculated for Expansion and Occasional Loads.

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INTERGRAPH

WRC 107 Stress Calculation for EXPansion loads:

Radial Load	P	1085.0	N.
Circumferential Shear	VC	4936.0	N.
Longitudinal Shear	VL	-1636.0	N.
Circumferential Moment	MC	3933.0	N.m.
Longitudinal Moment	ML	3031.0	N.m.
Torsional Moment	MT	8657.0	N.m.

Dimensionless Parameters used : Gamma = 48.03
Stress Concentration Factors Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction

Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
		Stress Values at (KPa)							
Circ. Memb.	P	-455	-455	-455	-455	-334	-334	-334	-334
Circ. Bend.	P	-820	820	-820	820	-1419	1419	-1419	1419
Circ. Memb.	MC	0	0	0	0	-3097	-3097	3097	3097
Circ. Bend.	MC	0	0	0	0	-37002	37002	37002	-37002
Circ. Memb.	ML	-6437	-6437	6437	6437	0	0	0	0
Circ. Bend.	ML	-10911	10911	10911	-10911	0	0	0	0
Tot. Circ. Str.		-18625	4839	16073	-4108	-41853	34989	38346	-32819
Long. Memb.	P	-334	-334	-334	-334	-455	-455	-455	-455
Long. Bend.	P	-1478	1478	-1478	1478	-815	815	-815	815
Long. Memb.	MC	0	0	0	0	-5312	-5312	5312	5312
Long. Bend.	MC	0	0	0	0	-18561	18561	18561	-18561
Long. Memb.	ML	-2296	-2296	2296	2296	0	0	0	0
Long. Bend.	ML	-15838	15838	15838	-15838	0	0	0	0
Tot. Long. Str.		-19947	14686	16321	-12397	-25144	13609	22603	-12889
Shear	VC	510	510	-510	-510	0	0	0	0
Shear	VL	0	0	0	0	169	169	-169	-169
Shear	MT	2765	2765	2765	2765	2765	2765	2765	2765
Tot. Shear		3276	3276	2255	2255	2935	2935	2596	2596
Str. Int.		22628	15676	18455	12971	42353	35385	38763	33152

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	1162.0	N.
Circumferential Shear	VC	107.0	N.
Longitudinal Shear	VL	-330.0	N.
Circumferential Moment	MC	-1175.0	N.m.
Longitudinal Moment	ML	1486.0	N.m.
Torsional Moment	MT	-416.0	N.m.

Dimensionless Parameters used : Gamma = 48.03
Stress Concentration Factors Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction

Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
		Stress Values at (KPa)							
Circ. Memb.	P	-487	-487	-487	-487	-358	-358	-358	-358
Circ. Bend.	P	-878	878	-878	878	-1520	1520	-1520	1520
Circ. Memb.	MC	0	0	0	0	925	925	-925	-925
Circ. Bend.	MC	0	0	0	0	11054	-11054	-11054	11054
Circ. Memb.	ML	-3156	-3156	3156	3156	0	0	0	0
Circ. Bend.	ML	-5349	5349	5349	-5349	0	0	0	0
Tot. Circ. Str.		-9872	2584	7139	-1801	10101	-8967	-13858	11291
Long. Memb.	P	-358	-358	-358	-358	-487	-487	-487	-487
Long. Bend.	P	-1583	1583	-1583	1583	-873	873	-873	873
Long. Memb.	MC	0	0	0	0	1587	1587	-1587	-1587
Long. Bend.	MC	0	0	0	0	5545	-5545	-5545	5545
Long. Memb.	ML	-1125	-1125	1125	1125	0	0	0	0
Long. Bend.	ML	-7764	7764	7764	-7764	0	0	0	0
Tot. Long. Str.		-10832	7864	6948	-5413	5771	-3572	-8493	4344
Shear	VC	11	11	-11	-11	0	0	0	0
Shear	VL	0	0	0	0	34	34	-34	-34
Shear	MT	-132	-132	-132	-132	-132	-132	-132	-132
Tot. Shear		-121	-121	-143	-143	-98	-98	-167	-167
Str. Int.		10847	7867	7216	5419	10104	8969	13863	11295

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WRC 107 Stress Summations:

Vessel Stress Summation at Attachment Junction

Type of Stress Int. / Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
	Stress Values at (KPa)							
Circ. Pm (SUS)	95596	95596	95596	95596	95596	95596	95596	95596
Circ. Pm (OCC)	0	0	0	0	0	0	0	0
Circ. Pm (TOTAL)	95596	95596	95596	95596	95596	95596	95596	95596
Circ. Pl (SUS)	-2	-2	137	137	-45	-45	144	144
Circ. Pl (OCC)	-3643	-3643	2668	2668	567	567	-1283	-1283
Circ. Pl (TOTAL)	-3646	-3646	2806	2806	521	521	-1138	-1138
Circ. Q (SUS)	2	-2	240	-240	-927	927	1348	-1348
Circ. Q (EXP)	-18625	4839	16073	-4108	-41853	34989	38346	-32819
Circ. Q (OCC)	-6228	6228	4470	-4470	9534	-9534	-12574	12574
Circ. Q (TOTAL)	-24850	11064	20784	-8819	-33246	26382	27120	-21594
Long. Pm (SUS)	46326	46326	46326	46326	46326	46326	46326	46326
Long. Pm (OCC)	0	0	0	0	0	0	0	0
Long. Pm (TOTAL)	46326	46326	46326	46326	46326	46326	46326	46326
Long. Pl (SUS)	24	24	74	74	-95	-95	230	230
Long. Pl (OCC)	-1483	-1483	767	767	1099	1099	-2074	-2074
Long. Pl (TOTAL)	-1459	-1459	842	842	1003	1003	-1843	-1843
Long. Q (SUS)	46	-46	391	-391	-450	450	692	-692
Long. Q (EXP)	-19947	14686	16321	-12397	-25144	13609	22603	-12889
Long. Q (OCC)	-9348	9348	6181	-6181	4672	-4672	-6418	6418
Long. Q (TOTAL)	-29248	23988	22894	-18970	-20922	9387	16876	-7162
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pm (OCC)	0	0	0	0	0	0	0	0
Shear Pm (TOTAL)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	-5	-5	5	5	218	218	-218	-218
Shear Pl (OCC)	11	11	-11	-11	34	34	-34	-34
Shear Pl (TOTAL)	5	5	-5	-5	252	252	-252	-252
Shear Q (SUS)	-247	-247	-247	-247	-247	-247	-247	-247
Shear Q (EXP)	3276	3276	2255	2255	2935	2935	2596	2596
Shear Q (OCC)	-132	-132	-132	-132	-132	-132	-132	-132
Shear Q (TOTAL)	2895	2895	1874	1874	2554	2554	2215	2215
Pm (SUS)	95596	95596	95596	95596	95596	95596	95596	95596
Pm (SUS+OCC)	95596	95596	95596	95596	95596	95596	95596	95596
Pm+Pl (SUS)	95594	95594	95734	95734	95552	95552	95742	95742
Pm+Pl (SUS+OCC)	91950	91950	98403	98403	96119	96119	94459	94459
Pm+Pl+Q (Total)	67262	103260	119259	89640	63086	122620	121642	72972

Circ. $P_m(SUS) = \frac{PD}{2T} = \frac{1.97*1844}{2*19} * 1000 = 95596$ kPa.

- Similarly we can calculate for Circ. P_m(OCC) if Occ. Pressure is Considered.

Circ. P_l(SUS)= Circ. Memb. P + Circ. Memb. M_l = 67+(-70)= -3 kPa

- Similarly we can calculate for Circ. P_l(OCC) as mentioned above.

Circ. Q (SUS) = Circ. Bend P + Circ. Bend M_l = 121+(-118)= 3 kPa

- Similarly we can calculate for Circ. Q(OCC) and Circ. Q(EXP) as mentioned above.

- **Use Same Philosophy to calculate Long. P_m, P_l, Q and Shear P_m, P_l, Q loads.**

Summation:-

Circ.P_m(Total)=P_m(SUS)+P_m(OCC)=95596 kPa

Circ.P_l(Total)= P_l(SUS)+P_l(OCC)= -3646 kPa

Circ.Q(Total)= Q(SUS)+Q(OCC)=-24850 kPa

P_m+P_l+Q (Total)= 95596+(-3646)+(-24850)= 67100 kPa

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Stresses in the Vessel at the Attachment Junction

Type of Stress Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
	Stress Values at (KPa)							
Pm (SUS)	95596	95596	95596	95596	95596	95596	95596	95596
Pm (SUS+OCC)	95596	95596	95596	95596	95596	95596	95596	95596
Pm+Pl (SUS)	95594	95594	95734	95734	95552	95552	95742	95742
Pm+Pl (SUS+OCC)	91950	91950	98403	98403	96119	96119	94459	94459
Pm+Pl+Q (Total)	67262	103260	119259	89640	63086	122620	121642	72972

Type of Stress Int.	Max. S.I. KPa	S.I. Allowable	Result
Pm (SUS)	95596	128465	Passed
Pm (SUS+OCC)	95596	154158	Passed
Pm+Pl (SUS)	95742	192697	Passed
Pm+Pl (SUS+OCC)	98403	231237	Passed
Pm+Pl+Q (TOTAL)	122620	399535	Passed

S_{mh}

1.2*S_{mh}

1.5*S_{mh}

1.5*1.2*S_{mh}

1.5*(S_{mh}+ S_{ch})

Max. Stress Obtained

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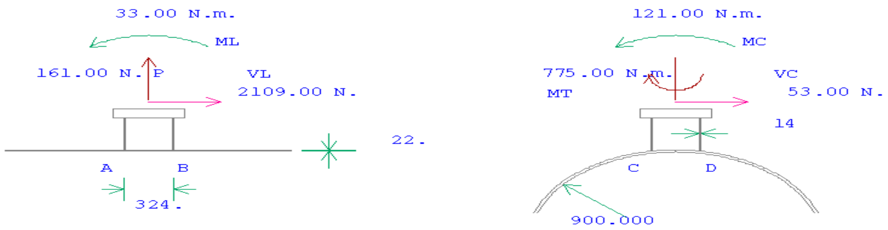
```
WRC 107 Analysis      :       New WRC 107(537
WRC 107 Version       :       MARCH, 1979, B1 & B2
Pressure               :       1.970  N./sq.mm.
Vessel Corr. Allow. :         3.0000
Nozzle Corr. Allow. :         3.0000
Load Input: SUStained Loads
```



Dimension Units :   mm.

WRC 297

Salient Features of WRC 297

- Extension of WRC107
- Used to determine local stresses at points on both Shell and Nozzle.
- Calculates loads in Radial, Longitudinal & Circumferential directions for both shell and nozzle.
- Thickness of nozzle is considered to calculate the loads at the junction.
- Loads are compared with the allowable corresponding to the materials of shell and nozzle

Limitations of WRC297

- Applied for cylindrical shell and nozzles connections only.
- $d/D < 0.5$,
- $20 \leq D/T \leq 2500$
- $d/t \geq 20$
- $d/T \geq 5$.
- For large D/T ratio the vessel is very flexible.

Hence flexibilities is used in CAESAR for large vessels.

FOREWORD

shallow-shell solutions used by Steele[3] may be valid up to a d/D ratio of 0.5, the exact limit of application depends on the D/T ratio. This is indicated by Figs. 3–58, where curves for different T/t and d/t values extend up to different values of λ. It is not recommended that curves be extrapolated beyond the λ values for which stress resultants are plotted.

In addition to the limitations given by Steele[3] denoted above, the limit D/T ≤ 2500 is recommended. The reason is that the theory is linear elastic and does not necessarily account for nonlinear effects or buckling. Test data are available for D/T up to 2500; those data give reasonable assurance that the theory is applicable up to D/T = 2500.

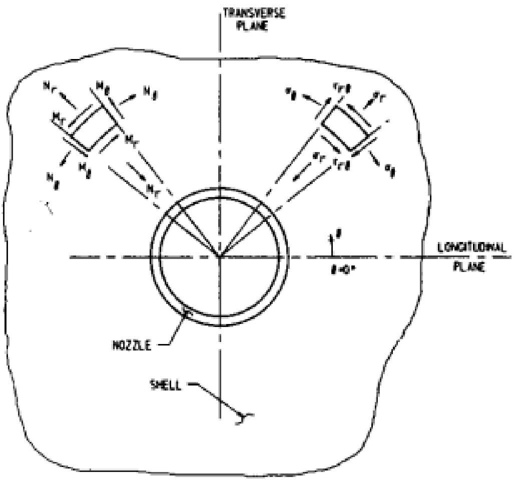
Limitations of WRC 297

- It is not recommended to use extrapolated values plotted obtained from curves.
- Applicable to 'isolated nozzles' having distance < 2 $\sqrt{DT}$ for Shell and < 2$\sqrt{dt}$ for Nozzle between the two nozzles.
- Nozzle must be attached to the vessel by through-penetration weld.
- Axial length of Nozzle should not be less than 2$\sqrt{dt}$.

The theory is applicable to "isolated" nozzles, where isolated means that the nozzle must be sufficiently remote from any other stress discontinuity so that the effect of the other discontinuity is negligible in the vicinity of nozzle-shell junction. Theoretically, this distance may be very large, but if the distance from the junction is greater than about $2\sqrt{DT}$ on the vessel or $2\sqrt{dt}$ on the nozzle, the theory is deemed to provide reasonable design guidance.

The theory is not applicable to nozzles that protrude inside the vessel. The nozzle, if of a different thickness than the attached branch pipe, should have an axial length of not less than $2\sqrt{dt}$. The nozzle must be attached to the vessel by a through-penetration weld.

WRC 297 Nomenclature



D	= mean diameter of vessel
d	= outside diameter of nozzle
T	= thickness of vessel
t	= thickness of nozzle
θ	= angle around nozzle (see Figs. 1 and 2)
λ	= $(d/D)(D/T)^{1/2}$
$P, M_c, M_L, M_T, V_c,$ and V_L	= nozzle loads as defined in Fig. 1
L	= length of vessel*
M_r, M_θ	= bending moments per unit length of shell wall (see Fig. 2)
N_r, N_θ	= membrane forces per unit length of shell wall (see Fig. 2)

Different Nomenclatures and ratio's are used to carry out for analysis when compared to WRC 107.

WRC 297 Input / Results & Comparison

INTERGRAPH

Identification	
Item Number	1
Description	New WRC 297/PD5500
Design	
PD5500 Annex G	False
Design Temperature	350
Design Pressure	1.970

Input to WRC297

For Comparison all Input parameters are kept same as that of WRC107 example.

Vessel	
Vessel Diameter Basis	OD
Vessel Diameter	1844.000
Wall Thickness	22.000
Corrosion Allowance	3.000
Stress Concentration Factor	1.000
Vessel Material	SA-516 70

Material of Vessel

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WRC 297 Input / Results & Comparison

INTERGRAPH

Nozzle/Attachment	
Reinforcing Pad	False
Nozzle Diameter Basis	OD
Nozzle Diameter	323.850
Wall Thickness	14.277
Corrosion Allowance	3.000
Stress Concentration Factor	1.000
Nozzle Material	SA-333 6

In WRC 197 global forces can be directly imported from the CAESAR file.

Pressure	
Include Pressure Thrust	False
Use Pressure Stress Indices (Div.2 AD 560.7)	False
Loads	
Radial Load (P)	1162.000
Circumferential Shear (VC)	4936.000
Longitudinal Shear (VL)	-2109.000
Torsional Moment (MT)	3933.000
Circumferential Moment (MC)	3031.000
Longitudinal Moment (ML)	8657.000

For entering the values of loads input direction with respect to vessel needs to be considered

Maximum loads are entered considering all cases of SUS, EXP & OCC

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WRC 297 Input / Results & Comparison

INTERGRAPH

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction

Type of Stress Int. / Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	92649	94621	92649	94621	92649	94621	92649	94621
Circ. Pl (SUS)	-19000	-19000	18078	18078	-3012	-3012	2516	2516
Circ. Q (SUS)	-28212	28212	26254	-26254	-35251	35251	31143	-31143
Long. Pm (SUS)	46324	46324	46324	46324	46324	46324	46324	46324
Long. Pl (SUS)	-5481	-5481	4985	4985	-4349	-4349	3427	3427
Long. Q (SUS)	-56514	56514	52406	-52406	-19297	19297	17339	-17339
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	510	510	-510	-510	213	213	-213	-213
Shear Q (SUS)	1254	1254	1254	1254	1254	1254	1254	1254
Pm (SUS)	92649	94621	92649	94621	92649	94621	92649	94621
Pm+Pl (SUS)	73656	75628	110731	112703	89637	91609	95166	97137
Pm+Pl+Q (Total)	61209	104282	136997	87554	54453	126892	126326	66026

Stress Values at (KPa)

Type of Stress Int.	Max. S.I. (KPa)	S.I. Allowable (KPa)	Result
Pm (SUS)	94621	128465	Passed
Pm+Pl (SUS)	112703	192697	Passed
Pm+Pl+Q (TOTAL)	136997	399535	Passed

Results of WRC 297 of Vessel

Stresses in the Vessel at the Attachment Junction

Type of Stress / Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Pm (SUS)	95596	95596	95596	95596	95596	95596	95596	95596
Pm (SUS+OCC)	95596	95596	95596	95596	95596	95596	95596	95596
Pm+Pl (SUS)	95594	95594	95734	95734	95552	95552	95742	95742
Pm+Pl (SUS+OCC)	91950	91950	98403	98403	96119	96119	94459	94459
Pm+Pl+Q (Total)	67262	103260	119259	89640	63086	122620	121642	72972

Stress Values at (KPa)

Type of Stress Int.	Max. S.I. (KPa)	S.I. Allowable (KPa)	Result
Pm (SUS)	95596	128465	Passed
Pm (SUS+OCC)	95596	154158	Passed
Pm+Pl (SUS)	95742	192697	Passed
Pm+Pl (SUS+OCC)	98403	231237	Passed
Pm+Pl+Q (TOTAL)	122620	399535	Passed

Results of WRC 107

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WRC 297 Input / Results & Comparison

INTERGRAPH

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction

Type of Stress Int. / Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30439	32804	30439	32804	30439	32804	30439	32804
Circ. Pl (SUS)	-19000	-19000	18078	18078	-4349	-4349	3427	3427
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15216	15216	15216	15216	15216	15216	15216	15216
Long. Pl (SUS)	-10451	-10451	10245	10245	-3722	-3722	3516	3516
Long. Q (SUS)	-113712	113712	106308	-106308	-74350	74350	66946	-66946
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	854	854	-854	-854	365	365	-365	-365
Shear Q (SUS)	2116	2116	2116	2116	2116	2116	2116	2116
Pm (SUS)	30439	32804	30439	32804	30439	32804	30439	32804
Pm+Pl (SUS)	11546	13883	48548	50910	26099	28462	33874	36238
Pm+Pl+Q (Total)	120532	118561	131788	131753	89084	85951	85737	84517

Stress Values at (KPa)

Type of Stress Int.	Max. S.I. (KPa)	S.I. Allowable (KPa)	Result
Pm (SUS)	32804	115426	Passed
Pm+Pl (SUS)	50910	173139	Passed
Pm+Pl+Q (TOTAL)	131788	349986	Passed

Loads of Nozzle

Allowable of Nozzle material

- Loads are calculated at 8 points on Shell as well as nozzle respectively.
- Results of Loads on vessel are obtained similar to WRC 107
- Thickness & material of Nozzle is used to calculate the loads.
- Range of ratio's is large for which calculation can be performed (e.g. 20 < D/T < 2500)

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Comparison of 107 and 297

WRC 107

- Calculates Local forces on shell in nozzle-shell connection
- Used for cylindrical and spherical shells
- Used for nozzles of rectangular, solid, round, hollow shapes
- Valid for d/D ≤ 0.33, D_m/T between 20-55
- Loads for various cases (SUS, EXP, OCC) can be separately entered

WRC 297

- Calculates Local forces on shell & nozzle in nozzle-shell connection
- Used for only for Cylindrical Shells
- Used for cylindrical nozzles only
- Used for d/D < 0.5 , 20 < D/T < 2500
- Maximum loads needs to be entered

Comparison of 107 and 297

WRC 107

- Considers weld factor
- Calculate loads for nozzle under fatigue
- Global element forces can directly imported form CAESAR File
- Calculates loads at 8 points on shell
- Curves range is less. Hence scope is limited
- Separate module is used is CAESAR to evaluate loads.

WRC 297

- Doesn't consider weld factor
- Doesn't calculate loads for nozzle under fatigue
- Manual input of loads considering vessel positioning
- Calculates loads at 16 points on shell & nozzles
- Wider range of curves are available. Hence covers curves outside the range and forms a supplement to WRC 107
- Nozzle flexibility is used is CAESAR to evaluate loads on nozzles, as nozzles are considered flexible

WRC 297 Flexibility

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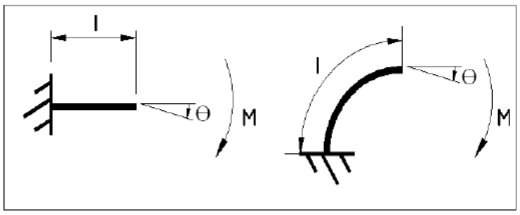
What is flexibility?

INTERGRAPH

International Journal of Modern Engineering Research (IJMER)
www.ijmer.com Vol. 4, Issue. 3, May-June. 2013 pp-3390-3397 ISSN: 2249-6645

Stress Intensification & Flexibility in Pipe Stress Analysis

Gaurav Bhende[1], Girish Tembhare[2]



A flexibility factor is defined as the rotation of a pipe bends that of straight pipe of same diameter, wall thickness and length under the action of the same bending moment. [3]

Example: for a bend

$k = 1.65 / h$ (2)

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Presented by GAURAV BHENDE in CAU EXPRESS 2017

WRC 297 flexibilities

- Nozzle loads on any vessel due to piping system needs to verified with the vendor allowables.
- Nozzles are considered to be flexible.
- Flexibility is evaluated as under:

Reference to NB-3686.5, Nozzle flexibility factor;

$$\theta = \frac{kMd}{EI_b}$$

Where;

θ= Rotational of nozzle axis with respect to vessel

k= stiffness

M= Moment due to loads in longitudinal and circumferential direction

d = Diameter of nozzle

E = Modulus of Elasticity

I_b = Moment of Inertia $=\pi d^3 t/8$

t = Thickness of Nozzle

WRC 297 flexibilities

Fig 60 provides values of $f(\lambda)=\frac{M}{ET^3\theta}$ corresponding to the values of λ & T/t

Where $\lambda = (d/D)\sqrt{D/T}$

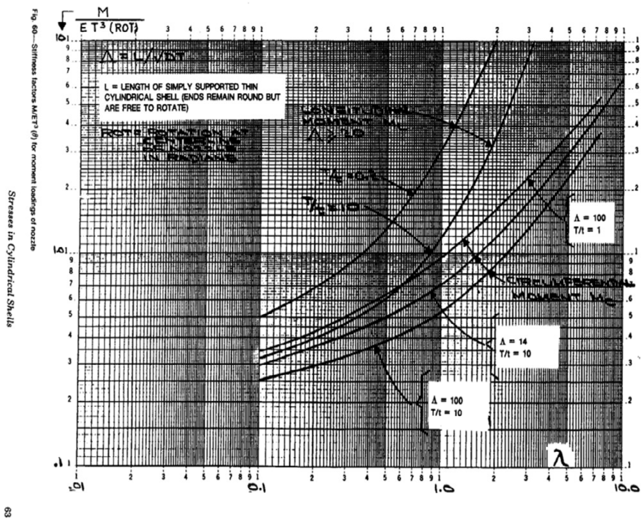
Equation can be written as

$$\theta = \frac{kMd}{EIb} = \frac{M}{ET^3 f(\lambda)}$$

$$k = \frac{Ib}{dT^3 f(\lambda)}$$

$$k = \frac{\pi}{8}\left(\frac{d}{t}\right)^2 \left(\frac{t}{T}\right)^3 \frac{1}{f(\lambda)}$$

Stiffness along longitudinal and circumferential axis can be evaluated using above formulae.



WRC 297 flexibilities

Radial loads are calculated by using fig 59

$\alpha = (P/w)/(P/w)_{PT.LOAD}$

Where,

α = axial stiffness

P = Radial nozzle load

w = Radial deflection due to P

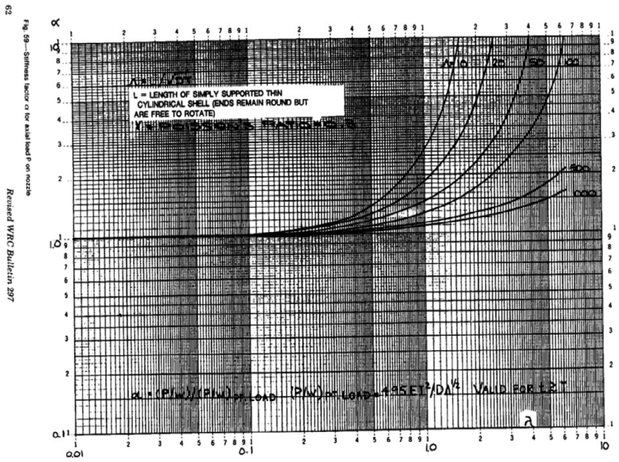
$(P/w)_{PT.LOAD} = 4.95\ ET^2/D\sqrt{\Lambda}$

Λ (Capital Lambda) $= L/\sqrt{DT}$

L = unsupported length of cylindrical shell

$= 8\ ab/\left(\sqrt{a} + \sqrt{b}\right)^2$

a, b = distance of nozzle from the stiffener points on the shell



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Calculation of WRC 297

Eg. D=1840mm, T=20mm, d=300NPD, t= 14.27mm, a=1050, b=900

$E = 2.0448 \times 10^{11}\ N/m^2$

$\lambda = (d/D)\sqrt{D/T} = (323.850/1820)\sqrt{1812/12}$

$= 2.196$

$L = 8\ ab/\left(\sqrt{a} + \sqrt{b}\right)^2$

$= 8 \times 1050 \times 900/\left(\sqrt{1050} + \sqrt{900}\right)^2$

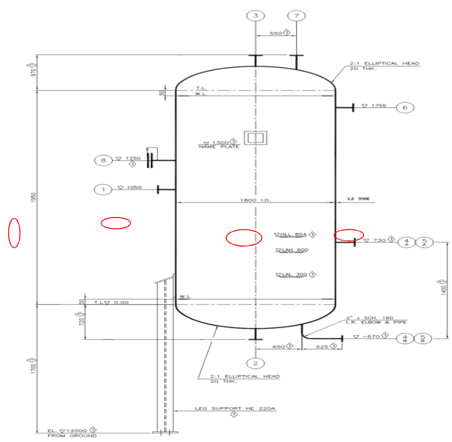
$= 1941.33mm$

$\Lambda = L/\sqrt{DT}$

$= 1941.33/\sqrt{1812 \times 12}$

$= 13.165$

$T/t = 12/14.27 = 0.840$



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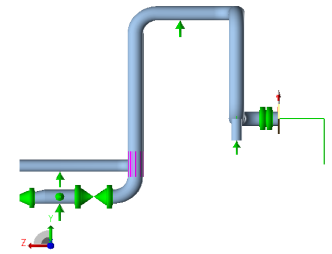
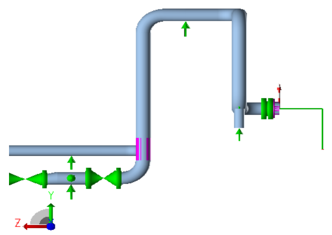
Calculation of WRC 297

INTERGRAPH

Node		fa N.	fb N.	fc N.	Forces Check	ma N.m.	mb N.m.	mc N.m.	Moments Check	
LOAD CASE DEFINITION KEY										
CASE 3 (OPE) W+T2+P1										
7050	Absolute Method									
	Limits	15000	15000	11000		11000	8500	9700		
	3(OPE)	5772	-3285	996	0.385	3045	-4912	-9889	1.019	*

Failure obtained





From fig. 60 for longitudinal Moments we have two values of f(λ) corresponding to values of T/t of 0.2 & 10 and λ= 2.196as 10 & 4 interpolating the values we have

$f(\lambda)_L = 9.608$ & $f(\lambda)_C = 1.68$

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Calculation of WRC 297

INTERGRAPH

From fig. 59 α= 10;

$K_L = f(\lambda) x\ ET^3 \ x\ \pi/180$

$= 6.416 \ x\ 2.0448x\ 1011 \ x\ (12/1000)3 \ x\ \pi/180$

$= 59229$ N.m/deg

Similarly values of K_C & α can evaluated by

$K_C = f(\lambda) x\ ET^3 \ x\ \pi/180 = 10360.512$ N.m/deg

$K_{\alpha x} = 4.95\ \alpha\ ET^2 / \sqrt{D^2\ \Lambda} = 2217340$ N/cm

Node		fa N.	fb N.	fc N.	Forces Check	ma N.m.	mb N.m.	mc N.m.	Moments Check
LOAD CASE DEFINITION KEY									
CASE 3 (OPE) W+T2+P1									
7050	Absolute Method								
	Limits	15000	15000	11000		11000	8500	9700	
	3(OPE)	4652	-3934	1322	0.310	2768	-1409	-6173	0.636

Within allowable limit

```
11  7050  WRC NOZZLE DATA FOR NODE 7050

          TERMINOLOGY:
          D1 - distance to stiffener or head
          D2 - distance to opposite side stiffener or head
          L  - unsupported length of cylindrical shell,
               L = 8(D1)(D2) / [ sqt(D1) + sqt(D2) ] ** 2
          D  - mean diameter of vessel
          T  - wall thickness of vessel
          d  - outside diameter of nozzle
          t  - wall thickness of nozzle
          &  - capital "lambda", & = L / sqt( DT )
          j  - "lambda", j = (d/D) * sqt(D/T)
          D1 =  1050.000   D2  =  900.000
          L  =  1941.338   T/t =    0.840
          D  =  1812.000   T   =   12.000
          d  =   323.850   t   =   14.278
          &  =    13.165   j   =    2.196

          CAUTIONS:
          Lambda (j) Greater than 2.0 produces approx. rigid
          junctions for longitudinal bending.
          The following combinations produce rigid axial junctions:
          j>1.6 &=10 / j>2.5 &=20 / j>4.0 &=50 / j>6.2 &=>100

12  7050  WRC 297  NOZZLE CALCULATIONS

               WRC NOZZLE NODE  = 7050
               VESSEL Dmean(mm.)= 1812.000     VESSEL THK. (mm.)=    12.000
               NOZZLE O.D. (mm.)=  323.850     NOZZLE THK. (mm.)=    14.278
          α    AXIAL TRANSLATIONAL STIFFNESS      ( N./cm.   )=     2216883
          KL   LONGITUDINAL    BENDING STIFFNESS (N.m./deg   )=       61669
               CIRCUMFERENTIAL BENDING STIFFNESS (N.m./deg   )=        8023
          KC   ANGLE BETWEEN NOZZLE & VESSEL CENTERLINES(deg)=       90.0000
               LENGTH (L) (mm.) = 1941.338     THICKNESS RATIO  =     0.840
               CAPITAL LAMBDA   =   13.165     SMALL LAMBDA     =     2.196
```

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